

PHYS 350: Honours Electricity and Magnetism

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Abstract

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1 Introduction

Textbook used is David J. Griffiths - Introduction to Electrodynamics Fifth Edition. Taught by Debanjan Sarkar & Adrian C. Liu. Mondays, Wednesdays and Fridays from 13:35 to 14:25 in rutherford 112. Course grading is divided as follows:

- 35% problem sets (must be handwritten scans).
- 25% midterm.
- 40% final exam.

2 Vector Analysis

2.1 Scalars, Vectors, and Fields

Definition 1 (Scalar). *A scalar is a quantity with just a magnitude*

Definition 2 (Vector). *Vector is a quantity with a magnitude and a direction. A vector can be written in the form of*

$$\vec{v} = v_x \hat{x} + v_y \hat{y} + v_z \hat{z}.$$

Definition 3 (Cross Product). *The cross product is a vector perpendicular to both, set by the right hand rule, and it vanishes for parallel vectors, and can be mathematically written as*

$$\vec{v} \times \vec{w} = |\vec{v}| |\vec{w}| \sin \theta \hat{n} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ v_x & v_y & v_z \\ w_x & w_y & w_z \end{vmatrix}.$$

Definition 4 (Dot Product). *The dot product measures how much of $\vec{v}$ points along $\vec{w}$, so it is zero for perpendicular vectors, and can be mathematically written as*

$$\vec{v} \cdot \vec{w} = v_x w_x + v_y w_y + v_z w_z = |\vec{v}| |\vec{w}| \cos \theta.$$

Remark 1. *Note $\vec{v} \times \vec{w} = -\vec{w} \times \vec{v}$, and $|\vec{v}| = \sqrt{\vec{v} \cdot \vec{v}}$.*

Definition 5 (Field). *A field is something with a value at every point in space. A scalar field assigns a number to each point, like temperature $T(x, y, z)$ or the height of a hill. A vector field assigns a vector to each point, like fluid velocity or $\vec{E}$.*

2.2 Divergence, Gradient, and Curl

Definition 6 (Del operator). *The del operator*

$$\vec{\nabla} = \left(\frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z} \right),$$

means nothing until it is fed a field.

$$\vec{\nabla} f = \frac{\partial f}{\partial x} \hat{x} + \frac{\partial f}{\partial y} \hat{y} + \frac{\partial f}{\partial z} \hat{z} \quad (\text{gradient: scalar} \rightarrow \text{vector}), \quad (1)$$

$$\vec{\nabla} \cdot \vec{F} = \frac{\partial F_x}{\partial x} + \frac{\partial F_y}{\partial y} + \frac{\partial F_z}{\partial z} \quad (\text{divergence: vector} \rightarrow \text{scalar}), \quad (2)$$

$$\vec{\nabla} \times \vec{F} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ F_x & F_y & F_z \end{vmatrix} \quad (\text{curl: vector} \rightarrow \text{vector}). \quad (3)$$

For the divergence we differentiate each component along its own direction. For the curl we mix directions, which is why terms like $\partial F_y / \partial x$ show up. The direction of the curl comes from the right hand rule: curl your fingers the way the paddle wheel turns and your thumb points along $\vec{\nabla} \times \vec{F}$.

The gradient points in the steepest uphill direction, and its length grows where the hill is steeper. It is always perpendicular to the contours of constant f .

2.3 Integrals

An integral is an instruction to move along a region, evaluate something at each step, and add it to a running total. The multivariable versions are:

$$\int_C \vec{F} \cdot d\vec{l} \quad \text{line integral, the work done if } \vec{F} \text{ is a force,} \quad (4)$$

$$\int_S \vec{F} \cdot d\vec{a} = \int_S \vec{F} \cdot \hat{n} da \quad \text{flux, how much of the field pokes through the surface,} \quad (5)$$

$$\int_V f d^3\vec{r} \quad \text{volume integral, the mass if } f \text{ is a density.} \quad (6)$$

We write $\oint$ when the curve is a closed loop or the surface is closed.

Example. Take $\vec{F} = y\hat{x}$ along a counterclockwise quarter circle of radius R . On the circle $d\vec{l} = R d\phi(-\sin\phi\hat{x} + \cos\phi\hat{y})$ and $y = R\sin\phi$, so

$$\int_C \vec{F} \cdot d\vec{l} = -R^2 \int_0^{\pi/2} \sin^2\phi d\phi = -\frac{R^2\pi}{4}.$$

It is negative because $\vec{F}$ points along $+\hat{x}$ while we travel in the $-\hat{x}$ direction over most of the arc.

2.4 The Three Fundamental Theorems

In single variable calculus, $\int_a^b (\partial f / \partial x) dx = f(b) - f(a)$. The same pattern holds in higher dimensions: integrating a derivative over a region reduces to evaluating the original object on the boundary of that region. The three theorems differ only in the dimension of the region.

Theorem 1 (Gradient theorem). *The boundary of a curve is two points:*

$$\int_a^b \vec{\nabla} h \cdot d\vec{l} = h(b) - h(a).$$

To find my altitude change climbing a hill, I can compare before and after, or add up all the little changes along the way.

Theorem 2 (Stokes' theorem). *The boundary of a surface is a curve:*

$$\int_S (\vec{\nabla} \times \vec{F}) \cdot d\vec{a} = \oint_C \vec{F} \cdot d\vec{l}.$$

To get the total swirl, I can work out the rotation at the boundary, or add up all the little whirlpools inside.

Theorem 3 (Gauss' divergence theorem). *The boundary of a volume is a surface:*

$$\int_V (\vec{\nabla} \cdot \vec{F}) d^3\vec{r} = \oint_S \vec{F} \cdot d\vec{a}.$$

To find the total flux leaking out through the surface, I can add up all the sources inside.

Remark 2. *Maxwell's equations come in a differential form written with $\vec{\nabla} \cdot$ and $\vec{\nabla} \times$, and an equivalent integral form written with $\oint \vec{F} \cdot d\vec{a}$ and $\oint \vec{F} \cdot d\vec{l}$. These two theorems are what let us pass between them. Gauss' law and Ampère's law are the first examples.*

2.5 Second Derivatives

Since we have three kinds of first derivative, we can mix and match to build second derivatives. Taking the divergence of a gradient gives

$$\vec{\nabla} \cdot (\vec{\nabla} f) = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} + \frac{\partial^2 f}{\partial z^2} \equiv \nabla^2 f,$$

where ∇^2 is the Laplacian. Mathematicians often write Δ , but in physics we use ∇^2 . Acting on a scalar field it returns a scalar field. It can also act on a vector field, component by component, returning a vector field.

The full list of second derivative identities:

$$\vec{\nabla} \cdot (\vec{\nabla} f) = \nabla^2 f \quad (\text{generally non-zero}), \quad (7)$$

$$\vec{\nabla} \times (\vec{\nabla} f) = 0 \quad (\text{curl of a gradient}), \quad (8)$$

$$\vec{\nabla} \cdot (\vec{\nabla} \times \vec{F}) = 0 \quad (\text{divergence of a curl}), \quad (9)$$

$$\vec{\nabla} \times (\vec{\nabla} \times \vec{F}) = \vec{\nabla}(\vec{\nabla} \cdot \vec{F}) - \nabla^2 \vec{F} \quad (\text{the BAC-CAB of vector calculus}). \quad (10)$$

These can be proved by writing out every partial derivative, which is tedious, or with index notation.

2.6 Index Notation

Write the i th component of $\vec{v}$ as v_i , so $v_1 = v_x$, $v_2 = v_y$, $v_3 = v_z$. A matrix carries two indices, A_{ij} , where i says which row and j says which column. A symmetric matrix has $A_{ij} = A_{ji}$, and an antisymmetric one has $B_{ij} = -B_{ji}$.

Definition 7 (Kronecker delta). *The identity matrix is written*

$$\delta_{ij} = \begin{cases} +1 & i = j \\ 0 & i \neq j \end{cases}.$$

Matrix multiplication $\vec{w} = A\vec{v}$ becomes $w_i = \sum_j A_{ij}v_j$. Multiplying by the identity gives $\sum_j \delta_{ij}v_j = v_i$, since the delta kills every term except $j = i$. This is the main trick: a delta summed against something sets one index equal to the other.

The dot product is $\vec{v} \cdot \vec{w} = \sum_i v_i w_i$, with no free index because the result is a scalar. The cross product needs a second symbol:

$$(\vec{v} \times \vec{w})_i = \sum_{j,k} \varepsilon_{ijk} v_j w_k.$$

Definition 8 (Levi-Civita symbol). ε_{ijk} is defined by

- $\varepsilon_{123} = +1$, and cyclic permutations keep the sign: $\varepsilon_{123} = \varepsilon_{231} = \varepsilon_{312}$.
- Swapping any two indices flips the sign, so $\varepsilon_{213} = \varepsilon_{132} = -1$.
- If any two indices are equal, $\varepsilon_{ijk} = 0$.

The identity that does all the work is

$$\sum_i \varepsilon_{ijk} \varepsilon_{imn} = \delta_{jm} \delta_{kn} - \delta_{jn} \delta_{km}, \quad (11)$$

which can be checked component by component.

Writing $\partial_i \equiv \partial/\partial x_i$, the derivatives become

$$\vec{\nabla} \cdot \vec{F} = \sum_i \partial_i F_i, \quad (\vec{\nabla} \times \vec{F})_i = \sum_{j,k} \varepsilon_{ijk} \partial_j F_k.$$

2.7 Proving the Identities

Set $\vec{G} \equiv \vec{\nabla} \times (\vec{\nabla} \times \vec{F})$. Then

$$G_i = \sum_{j,k} \varepsilon_{ijk} \partial_j \left(\sum_{m,n} \varepsilon_{kmn} \partial_m F_n \right) \quad (12)$$

$$= \sum_{j,m,n} \left(\sum_k \varepsilon_{kij} \varepsilon_{kmn} \right) \partial_j \partial_m F_n \quad (13)$$

$$= \sum_{j,m,n} (\delta_{im} \delta_{jn} - \delta_{in} \delta_{jm}) \partial_j \partial_m F_n. \quad (14)$$

In the second line we used the cyclic property $\varepsilon_{ijk} = \varepsilon_{kij}$ to move the summed index to the front so the identity applies. The first term gives

$$\sum_j \partial_j \partial_i F_j = \partial_i \left(\sum_j \partial_j F_j \right) = \left(\vec{\nabla} (\vec{\nabla} \cdot \vec{F}) \right)_i,$$

and the second gives $-\sum_j \partial_j \partial_j F_i = -(\nabla^2 \vec{F})_i$. Together they reproduce the identity claimed above.

For the curl of a gradient,

$$\left(\vec{\nabla} \times (\vec{\nabla} f) \right)_i = \sum_{j,k} \varepsilon_{ijk} \partial_j \partial_k f. \quad (*)$$

Now j and k are internal to the sum, so their names do not matter. Swap the labels:

$$= \sum_{j,k} \varepsilon_{ikj} \partial_k \partial_j f = - \sum_{j,k} \varepsilon_{ijk} \partial_k \partial_j f = - \sum_{j,k} \varepsilon_{ijk} \partial_j \partial_k f. \quad (**)$$

The second step uses the antisymmetry of ε and the third uses the fact that partial derivatives commute. Comparing (*) and (**), the expression equals minus itself, and the only such number is zero. So $\vec{\nabla} \times (\vec{\nabla} f) = 0$.

2.8 Irrotational and Solenoidal Fields

The two vanishing identities let us classify fields.

- Since $\vec{\nabla} \times (\vec{\nabla} \psi) = 0$, writing $\vec{F} = \vec{\nabla} \psi$ guarantees $\vec{\nabla} \times \vec{F} = 0$. Such a field is **irrotational**.
- Since $\vec{\nabla} \cdot (\vec{\nabla} \times \vec{A}) = 0$, writing $\vec{F} = \vec{\nabla} \times \vec{A}$ guarantees $\vec{\nabla} \cdot \vec{F} = 0$. Such a field is **solenoidal**.

Any vector field can be written as $\vec{\nabla} \psi + \vec{\nabla} \times \vec{A}$.

Theorem 4 (Irrotational fields). *The following four statements are equivalent:*

- (i) $\vec{\nabla} \times \vec{F} = 0$ everywhere.
- (ii) $\int_a^b \vec{F} \cdot d\vec{l}$ is independent of path.
- (iii) $\oint_C \vec{F} \cdot d\vec{l} = 0$ for any closed loop.
- (iv) $\vec{F} = \vec{\nabla}\psi$ for some scalar ψ .

Theorem 5 (Solenoidal fields). *The following four statements are equivalent:*

- (i) $\vec{\nabla} \cdot \vec{F} = 0$ everywhere.
- (ii) $\int_S \vec{F} \cdot d\vec{a}$ is independent of the surface, for surfaces sharing a boundary.
- (iii) $\oint_S \vec{F} \cdot d\vec{a} = 0$ for any closed surface.
- (iv) $\vec{F} = \vec{\nabla} \times \vec{A}$ for some vector field $\vec{A}$.

Remark 3. *In electrostatics we will find $\vec{\nabla} \times \vec{E} = 0$. That is the irrotational case, and it is what licenses us to write $\vec{E} = -\vec{\nabla}V$ and introduce the electric potential. In magnetostatics we will find $\vec{\nabla} \cdot \vec{B} = 0$, which is the solenoidal case and licenses $\vec{B} = \vec{\nabla} \times \vec{A}$, the vector potential.*

3 Electrostatics

4 Potentials

5 Electric Fields in Matter

6 Magnetostatics

7 Magnetic Fields in Matter

8 Electrodynamics

A Appendix

B Useful Links